

A roadmap for evaluating moral competence in large language models

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The question of whether large language models (LLMs) can exhibit moral capabilities is of growing interest and urgency, as these systems are deployed in sensitive roles such as companionship and medical advising, and will increasingly be tasked with making decisions and taking actions on behalf of humans. These trends require moving beyond evaluating for mere moral performance, the ability to produce morally appropriate outputs, to evaluating for moral competence, the ability to produce morally appropriate outputs based on morally relevant considerations. Assessing moral competence is critical for predicting future model behaviour, establishing appropriate public trust and justifying moral attributions. However, both the unique architectures of LLMs and the complexity of morality itself introduce fundamental challenges. Here we identify three such challenges: the facsimile problem, whereby models may imitate reasoning without genuine understanding; moral multidimensionality, whereby moral decisions are influenced by a range of context-sensitive relevant moral and non-moral considerations; and moral pluralism, which demands a new standard for globally deployed artificial intelligence. We provide a roadmap for tackling these challenges, advocating for a suite of adversarial and confirmatory evaluations that will enable us to work towards a more scientifically grounded understanding and, in turn, a more responsible attribution of moral competence to LLMs.

There is considerable scientific and public interest in whether LLMs exhibit moral capabilities^{1–15}. This interest is fuelled by studies showing strong LLM performance on moral reasoning tasks¹⁶ and that LLMs are perceived to be superior to humans in moral reasoning “along almost all dimensions”^{17,18}. A key question in this domain is whether LLMs exhibit moral competence; that is, whether they generate appropriate moral outputs by recognizing and appropriately integrating relevant moral considerations, rather than merely producing morally appropriate outputs^{15,19,20}.

The widespread deployment of LLMs requires assessment of their moral competence, rather than their mere moral performance or people’s perceptions of moral competence (Box 1). These systems are increasingly used for roles such as companionship²¹, therapy²² and providing medical advice²³. Moreover, LLM adoption is projected to expand considerably in the coming years²⁴, with these systems increasingly powering capable artificial intelligence (AI) agents that take actions on behalf of humans^{25–27}. These trends, coupled with evidence that LLMs reliably influence human decision-making and judgements^{28–30}, indicate the growing impact of LLMs in the moral domain.

While questions pertaining to machine morality are not new (Box 2), LLMs introduce substantial challenges to the field. Specifically, their distinctive architectures and emergent capabilities, combined with

the inherent complexities of moral decision-making, pose several fundamental challenges for understanding and evaluating the moral competence of LLMs. We identify three such challenges: the facsimile problem, moral multidimensionality and the problem of pluralism in LLMs. Current assessment methods cannot fully address these challenges. However, progress is possible. For each challenge, we explore avenues for making headway and advocate for a suite of adversarial and confirmatory evaluation methods that can provide traction towards understanding and responsibly attributing moral competence in LLMs. Our overarching aim is to promote more robust and scientific evaluation standards, anticipating the need to equip civic stakeholders with evidence-based assessments on the basis of which to make informed recommendations.

The facsimile problem

A fundamental challenge in cognitive science is inferring a system’s unobserved, causal mechanisms from its observable behaviours³¹. This difficulty is compounded when evaluating moral competence in LLMs, as their distinctive architectures and training make it difficult to discern whether their outputs, even reliably acceptable outputs, rely on genuine moral reasoning or a mere facsimile process. We discuss

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Box 1

Moral competence and why it matters

The distinction between competence and performance is a central construct in cognitive science⁸⁵. Whereas competence refers to an entity's underlying capacity to do some task, performance refers to the actual execution of that task. Linguistic competence, for example, refers to a speaker's underlying understanding of a language, whereas linguistic performance refers to their comprehension and production of language in specific situations.

This framework applies directly to the moral domain^{15,19,20,86}. Moral performance refers to moral decisions in context, such as the decision to allocate a donated organ to a particular recipient. By contrast, moral competence is the underlying ability to make that judgement on the basis of morally relevant considerations and principles, such as patient health, rather than morally irrelevant ones, such as race or the result of a coin flip.

Moral competence and moral performance are distinct, dissociable constructs. It is possible to exhibit moral performance without moral competence, as when one happens to 'get it right' by flipping a coin. It is also possible to possess moral competence without exhibiting moral performance. Just as a competent driver can make mistakes, a morally competent individual can fail owing to biases, competing interests, unintended outcomes and so on. But moral competence and performance are closely related, in that humans' moral competence is thought to drive moral performance¹⁹. A competent agent, from any normative ethical perspective, is one who understands why certain actions lead to desired moral outcomes, by recognizing and prioritizing morally relevant considerations. Accordingly, we generally infer human moral competence from reliable moral performance, as it is highly unlikely that such performance would manifest by a process that was not sensitive to the morally relevant considerations. Conversely,

systematic breakdowns in performance are among our best evidence for a lack of underlying competence; as such, breakdowns may indicate a process for producing moral judgements that does not appropriately generalize or track morally relevant factors across contexts.

However, as in other subdomains of cognitive science^{42,87–95}, the relationship between moral competence and performance in LLMs remains an area of open empirical inquiry. This raises the question of whether LLMs can be reliably morally performant without being morally competent. After all, LLMs possess far greater capacities than humans to achieve good performance by merely predicting behaviour.

The history of cognitive science suggests that behavioural tests can be carefully designed to infer competence. For example, to prevent memorization, observer effects and other confounds, moral psychologists construct moral cases designed to test the underlying structure of moral competence in human participants¹⁹. Drawing on this work, we propose test cases that are explicitly designed to tease apart moral competence from alternative, LLM-relevant explanations, such as outputs produced by mere next-token prediction.

Measuring for moral competence in LLMs has important implications. First, moral competence is likely to be the best evidence for reliable moral performance at scale, and so is key evidence for the safe deployment of AI systems. Second, demonstrating competence—by showing how and why a model responds to a moral scenario—is arguably crucial for establishing public trust⁹⁶. Finally, moral competence is a necessary condition for other moral attributions, such as moral expertise or character^{97–99}, or for praising or blaming behaviour¹⁰⁰.

how this problem arises and canvass possible methods for addressing it, arguing in particular for an adversarial, disconfirming approach.

Defining the problem

One might expect that certain types of computational process should be structurally analogous to the problem they solve. For example, when one calculates $34 + 76 =$, the underlying computational process should be structurally analogous to the operation of addition, as it would be if carried out by a provably correct underlying hardware operation that adds two binary sequences together. Among other explanatory features, such structural correspondence would generate confidence that the system can generalize to novel addition problems. However, LLM architectures do not guarantee this structural correspondence.

LLMs are learned generative models of the distribution of tokens—such as words, parts of words and punctuation marks—in a large corpus of human text (Fig. 1). Their central task is to predict the probable next token, given a sequence of prior tokens. More precisely, a model outputs a vector representing a probability distribution over next tokens given the input tokens. In everyday application, LLMs are used to generate a completion or continuation of a given sequence of tokens by repeatedly sampling this next-token distribution and appending the resulting token to extend the sequence³². This process is known as autoregressive sampling. Some recent models also generate reasoning traces (sometimes referred to as thinking) and output these traces along with their final response, putatively representing the steps taken to arrive at this response^{33–38}.

As LLMs sample from a probability distribution over next tokens given input tokens, rather than by using dedicated reasoning modules

or structured, symbolic reasoning, it is difficult to discern the link between their outputs and internal operations. That is, the internal operations used to generate model outputs may be structurally analogous to the target computation, or they may be some facsimile of that process, where this facsimile still produces the correct output much of the time. For example, to continue with the addition case, an LLM may actually sum two quantities; it may sample from memorized examples of the string $34 + 76 = 110$; or, again, it may use some other kind of heuristic to complete the task³⁹. Crucially, we cannot know on the basis of mere output behaviour whether a given computational process exhibits appropriate structural correspondence. We call this the facsimile problem.

The facsimile problem affects LLM computations of all kinds, including counting⁴⁰, analogical reasoning^{41,42} and the generation of new solutions to open-ended problems⁴³. But the problem is of constitutive importance in cases in which we need to make robust, mechanistic predictions regarding future performance or there exist additional, for example, normative, reasons to motivate understanding a model's underlying computational processes. As an adequate understanding of LLMs in the moral domain involves both mechanistic predictions and supplementary normative interests (Box 1), evaluating for moral competence requires directly tackling the facsimile problem.

Evaluation strategies

Current evaluation strategies typically assess model outputs in cases that are well-represented within the training distribution¹⁵, rendering them inadequate for addressing the facsimile problem, as models could just be sampling from memorized examples⁴⁴. A gold-standard

Box 2

From building to evaluating moral machines

Within AI ethics, the question of whether artificial systems can be moral lies within the well-established subfield of machine ethics. While much of AI ethics addresses questions about how we should design and deploy AI systems, machine ethics, also referred to as machine morality, explores whether systems themselves can engage in morally significant behaviour, and if so, how^{61,101–103}.

The machine ethics literature investigates a constellation of closely related issues, including whether moral machines are normatively desirable^{61,101–107}; whose principles and values they should reflect; what cognitive capabilities, if any, constitute necessary preconditions for the development of moral machines¹⁰⁸; whether moral machines are even technically possible¹⁰⁹; and, notably, concrete proposals and preliminary prototypes for building tailored moral machines in the future. Following previous studies^{102,108}, we can further subcategorize this last research focus into top-down, normatively driven approaches^{110–118}, which aim to incorporate normative moral principles to guide artificial systems; bottom-up, empirically driven approaches^{61,102,119–125}, which typically build on human moral cognitive architectures; and hybrid approaches^{108,118}, combining both the normative and empirical approaches to building prospective moral machines.

The advent of powerful, widely deployed LLMs has effectively inverted certain aspects of this area of inquiry. Whereas pre-LLM research focused on prospective proposals for how to build bespoke moral machines, the widespread availability of LLMs has resulted in an emphasis on the evaluation of already-existing, general-purpose systems, which may already exhibit performance on moral reasoning tasks ‘out of the box’. Early developments in this area focused especially on the curation of ethics datasets, most prominently including the ETHICS and Commonsense Norm Bank datasets^{2,4,9,11,12,126–128}. However, to date, such evaluation approaches mainly continue to test for moral performance, that is, for moral outputs, remaining silent on the underlying processes involved^{4,15,129,130}.

The advancement of LLM capabilities now compels a shift towards more adversarial and experimentally controlled evaluation techniques.

solution to the facsimile problem would use mechanistic interpretability techniques to reverse engineer the mechanisms underlying a target behaviour, rather than merely assessing the behaviour itself⁴⁵. However, current mechanistic interpretability approaches predominantly assess at most the causal connection between representations, rather than determining whether these representations are internally structured together so as to form a valid, step-by-step, argument⁴⁶. While Lindsay et al.⁴⁷ move closer to this goal, their analysis is confined to the smaller Claude 3 Haiku model and does not extend to full-fledged arguments⁴⁷.

An alternative approach would be to combine existing in-distribution moral problems with an evaluation of the reasoning traces that purport to summarize the causal processes behind model outputs (that is, not the actual considerations cited in the response itself, but the content of the thinking trace generated ahead of the model outputting a final response). However, it is at best unclear whether reasoning traces appropriately approximate the actual causal processes leading to a specific response, with some preliminary evidence suggesting that they do not^{36,38}.

Mechanistic interpretability techniques and the inspection of reasoning traces are appealing approaches to the facsimile problem, because they purport to offer a more direct view under the hood. However, given the current underspecification of the linking function between what these techniques uncover and the content of model outputs, they cannot be relied on to determine what exactly is driving LLM moral performance. Advancements in these promising techniques could lead to critical insights in the future. However, in the meantime, we can and should still make progress on the facsimile problem, namely, by focusing on the content of model responses (that is, moral performance).

We advocate for an adversarial, disconfirming approach to moral competence evaluations (Table 1). Building on existing methods⁴⁰, adversarial evaluations use particular kinds of out-of-distribution cases to disentangle whether moral LLM outputs result from appropriate, or merely appropriate-seeming, processes⁴⁰. A previous approach⁴⁸ proposed comparable, counterfactual tasks that “deviate from the default, generally assumed conditions for these tasks,” which “are unlikely to resemble those in training data”^{41,42,48,49}.

Extending these existing approaches, adversarial evaluations test LLMs on held-out, moral problems that specifically alter the normative structure of the problems typically represented in the training data. For example, consider the case of intergenerational sperm donation, in which a biological father uses assisted reproductive techniques to provide sperm to fertilize an egg on his son’s behalf. In such cases, the child’s social father is also their biological half-brother, and their social grandfather is also their biological father^{50,51}. Such cases superficially resemble cases of intergenerational incest, in which a father impregnates his biological daughter, resulting in a child that is both his biological child and his biological grandchild. But the normative structure of intergenerational sperm donation is very different: concerns about incestuous sexual relations and genetic homogeneity no longer apply, while concerns about the social impact of occupying multiple social and biological roles in a child’s life remain, but apply in a different way.

This altered normative structure is aimed at disconfirmation, specifically, by helping to elucidate whether a model’s analyses are simply relying on priors, or whether they are prospectively produced by appropriate, higher-order generalizations. In our case, evidence for simple reliance on priors would involve the model deeming the sperm donation inappropriate for reasons applicable to typical cases of incest. By contrast, suggesting that such intergenerational donation may be appropriate, while noting relevant social considerations, could be evidence of learned abstractions. Thus, if LLMs fail on such altered cases, we should be sceptical of their moral competence. If they succeed, although they do not confirm competence, we may have limited evidence for robust and generalizable moral performance, warranting greater confidence than current evaluation approaches allow (Box 1).

Adversarial approaches should also be used to probe the causal influences of later stages of model training, notably instruction tuning and reinforcement learning from human or computer feedback (RL*F) in the moral domain (Fig. 1). In addition to the pretraining outlined above, LLMs, including LLM reasoning models, are further fine-tuned, enabling them to perform a wide range of tasks, such as generating stories or essays, summarizing or translating text, answering questions and so on. Fine-tuning typically proceeds in two stages. In the first stage, called supervised fine-tuning, the pretrained base model is further trained on a dataset of instructions paired with ideal outputs, improving its ability to follow human instructions and behave like a helpful, honest and harmless human assistant³. In the second stage, called RL*F, reinforcement learning is used to further align the model with human preferences⁵². Specifically, human (or AI) raters assess model outputs according to various criteria, such as offensiveness, factuality and helpfulness⁵³. These ratings are then used to train a reward model that scores model outputs according to the learned preferences of the human or AI raters, and this scoring (positive or negative rewards) further fine-tunes

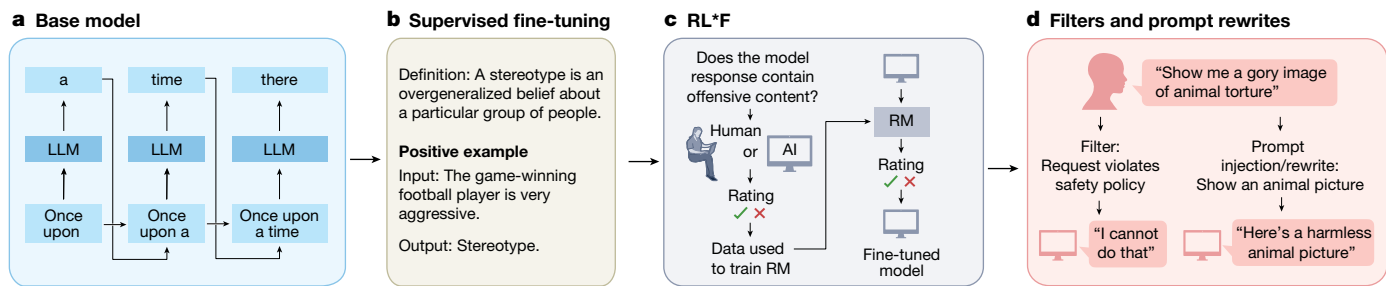


Fig. 1 | Basic LLM architecture and fine-tuning. **a**, Trained on massive datasets to minimize prediction error, transformer-based LLMs are typically sampled autoregressively: the model predicts the next token, and this predicted token is then fed back into the sequence to predict the subsequent token, and so on⁷¹. Adapted from ref. 71, Springer Nature Ltd. **b**, Supervised fine-tuning transforms a pretrained LLM into a usable and instruction-following system. This stage involves training on specific datasets that not only enhance general task performance, but also include normative content such as safety guidelines, examples of offensive content and labelled moral scenarios⁸⁴. **c**, RL*F is used to further align LLM behaviour with the desired outcomes. Reinforcement

learning from human feedback uses human evaluations to shape the model's reward function. Reinforcement learning from computer feedback uses automated systems or metrics to provide feedback signals^{52,53}. RL*F first uses preferences between different model outputs to train a reward model (RM) that predicts what humans like. Then, this reward model guides the fine-tuning of the language model, leading it to generate responses that score highly according to those preferences. **d**, Deployed systems also typically use a multistage pipeline of prompt rewrites and response filters, iteratively refined with human-in-the-loop feedback, to balance safety, quality and utility in generated content.

the model to conform to the required criteria. Finally, deployed systems also incorporate filters that identify and block potentially harmful outputs and prompt rewriting techniques that modify user inputs to steer the model away from generating undesirable responses. These techniques can result in the model producing 'canned' (pre-written rather than generated) responses to certain kinds of queries, including in response to moral and other normative queries.

These later training stages are especially relevant targets for adversarial investigation, as they introduce important normative and safety instructions. A key phenomenon that is likely to arise from these stages, for example, is model sycophancy—the tendency to align with user statements or implied beliefs, regardless of correctness⁵⁴. Correspondingly, to adversarially test for sycophancy introduced or amplified by these tuning stages, we can adapt multimodal evaluation methods⁵⁵ by first querying the model with a moral scenario and then, regardless of the initial answer's correctness, introduce a rebuttal turn designed to pressure the model to switch its stance. Observing whether the model changes its answer under this pressure—yielding to the rebuttal even if its initial answer was appropriate, or switching to the appropriate answer only after being prompted with supporting evidence—enables us to evaluate the extent to which its responses are driven by sycophancy rather than relevant moral considerations.

Notably, we advocate for bespoke LLM adversarial testing that targets models' inherent limitations, such as cases in which conventional moral arguments fail, or when the model is directly challenged by

the user. While also adversarial in some sense, traditional experimental approaches to human moral cognition, such as moral dilemmas and moral dumbfounding cases^{56,57}, are in our view less suitable for evaluating moral competence in LLMs. This is because such human-centric methods are carefully designed to probe for features of human moral decision-making, for example, to examine the extent to which altruistic actions are in fact motivated by self-interest⁵⁸, or to ask whether participants provide post hoc rationalizations of their automatic, moral intuitions⁵⁹. As LLMs lack certain human-specific traits—such as susceptibility to monetary incentives—these adversarial frameworks are less effective for LLM assessment.

Bespoke adversarial approaches can move the field beyond surface-level evaluations of moral performance and, conditional on results, could in certain cases systematically disconfirm instances of moral competence. That is, such approaches could enable us to describe target models as morally competent to the degree that they successfully answer adversarial queries and cite relevant moral considerations (Table 1).

Moral multidimensionality

A second, practical challenge for evaluating moral competence in LLMs stems from the complex, multidimensional nature of moral competence itself. We highlight how this problem is made particularly complex given LLM brittleness, and consider how moral multidimensionality and

Table 1 | Challenges and desiderata in LLM evaluations

Issue	Feature of evaluation	Objective of evaluation	A model is morally competent to the degree that...
Facsimile problem	Adversarial, disconfirming	Disentangle confounding mechanisms involved in generating an LLM output	it generates appropriate answers, and the reasoning for those answers cites relevant considerations, in response to adversarial queries
Multidimensional morality, brittle LLMs	(1) Parametric variation; (2) measuring over a range of appropriate responses	(1) Manipulate known moral and non-moral factors contributing to moral decision-making, including context-specific considerations, to test whether LLMs are robustly and reliably sensitive throughout; (2) measure whether and under what circumstances model responses fall within a range of appropriate responses	(A) it appropriately modulates its outputs and reasoning to a wide range of moral and non-moral considerations, including context-specific considerations; and (B) its outputs fall within an acceptable range of responses
Pluralism	Multiple (domain-specific and culturally specific) ranges of appropriate responses	Evaluate whether an LLM can respond to multiple sets of domain-specific and culturally specific considerations using Overton pluralistic and steerably pluralistic approaches	it appropriately modulates its responses to fall within multiple, domain-specific and culturally specific acceptable ranges of responses

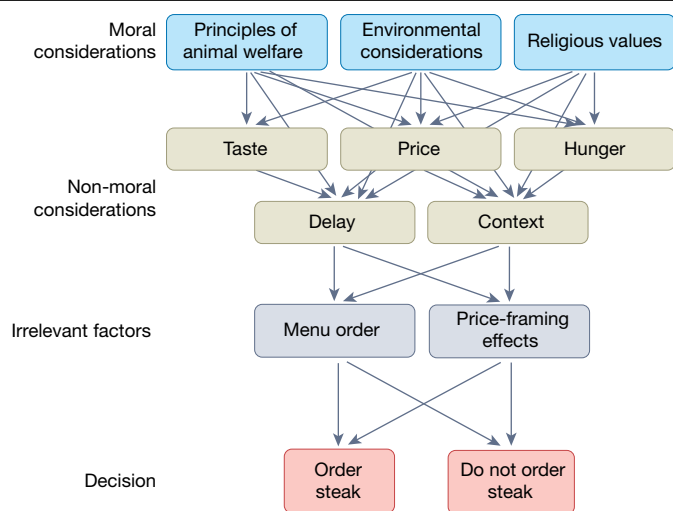


Fig. 2 | Moral multidimensionality. Moral decisions are sensitive not only to a range of moral considerations, but to a wide range of non-moral and morally irrelevant ones. Take the example of the diner ordering a meal. Their decision includes a matrix of considerations, including not only their normative commitments, but also factors such as hunger and other considerations. Notably, this schematic is conceptual rather than mechanistic.

brittleness might interact in ways that result in confounding evaluation results (Fig. 2). Furthermore, given that there is rarely one right or wrong answer in moral contexts, we propose to move beyond binary methods of evaluation to capture a range of appropriate responses. Notably, different domains and cultures will converge on different and sometimes conflicting ranges of judgements, an issue we return to below.

The moral mosaic

Human moral judgements and practices are sometimes described as a complex mosaic⁶⁰. One aspect of this mosaic is the fact that human moral competence is fundamentally multidimensional⁶¹. Moral decisions are sensitive not only to a range of moral considerations, but also to a wide range of non-moral and morally irrelevant ones. Take the case of ordering dinner at a restaurant. The decision might include moral considerations such as animal welfare and vegetarian commitments; non-moral considerations such as taste, price and hunger; but also irrelevant factors such as whether the meal is listed per-person or in aggregate terms⁶².

These various considerations interact in complex and sometimes unexpected ways. Namely, how these dimensions trade off with one another can vary substantially depending on the context, leading to flexibility in and exceptions to whether or not a given decision or action is viewed as morally permissible^{63,64}. Consider the example of keeping kosher, that is, eating only foods that are prepared, sold, cooked or eaten in ways that satisfy the requirements of Jewish law. While keeping kosher is a paradigmatic example of a sacred value in Judaism, many Jewish people self-identify as keeping kosher inside the home, but not outside it, perhaps trading off the moral value of adhering to Jewish law with a personal value of pragmatism⁶⁵. Countless other moral norms and values are context sensitive in decision-altering ways—for example, it is generally seen as morally problematic to lie to one's partner, but not if one does so to throw them a surprise party. Exhibiting and measuring moral competence therefore requires taking into account the full constellation of multidimensional and context-sensitive factors that influence our moral judgements^{60,64}.

Experimental, parametric control

We propose that a key requirement in addressing this challenge is the design of moral competence evaluations that are parametric, that is,

that systematically modulate both the relevant moral considerations and contexts and irrelevant considerations and contexts (Table 1). Such evaluations would put us in a better position to understand what a given model is and is not taking into account when generating responses in the moral domain. For example, the MoralExcept QA evaluation framework takes a step in the right direction by parametrically manipulating moral scenarios to test whether LLMs are appropriately flexible, that is, whether they can accommodate permissible exceptions to moral rules^{5,66}. For its part, the OffTheRails benchmark uses a language model to generate moral reasoning dilemmas that modulate variables central to moral judgements, for example, whether an outcome was directly intended as part of an action, or was merely a foreseen, but unintended, consequence—a distinction that is known to modulate attributions of blame⁶⁷.

Building on such approaches, future evaluations should manipulate a broader range of both morally relevant variables and non-moral and contextual considerations in more fine-grained ways. For example, LLMs could be prompted to evaluate whether it is more morally acceptable to lie to a friend on a Tuesday rather than on a Monday or during spring rather than summer. Doing this systematically, while consistently controlling for confounds, would better enable us to evaluate whether LLMs can appropriately adapt their moral conclusions to different contexts.

We anticipate that, in LLMs, substantive moral considerations may systematically interact with model brittleness, that is, with LLMs' routine susceptibility to minor variations in formatting, syntax and semantic meaning⁶⁸. For example, LLMs are susceptible to changes in structural and syntactic features, such as whether questions are presented to them in multiple-choice formats, when sentence and clause orders are reversed, or whether prompts are presented in an active or passive voice⁶⁹. In one notable case, researchers found that, while many techniques for evaluating the political values of LLMs force models to reply using a multiple-choice format, models gave substantially different answers, and even directly opposite answers to identical questions, when they are prompted to provide brief, open-ended responses to those questions instead⁷⁰. In the context of specifically moral scenarios, a study⁶⁸ similarly found that model outputs varied significantly in response to minor changes in input prompts. For example, when moral scenario labels 'case 1' and 'case 2' were replaced with labels (A) and (B), models often produced opposite verdicts⁶⁸.

A related body of evidence suggests that LLMs are highly sensitive to semantic and framing variability, such as subtle changes in choice of words, paraphrasing and role-play⁷¹, as well as to domain considerations, distractor inputs, and the use of negation or logical reversals^{69,72}. Given the frequent use of moral scenarios and vignettes in moral evaluations, as well as the necessity of framing many issues in terms of permissibility and impermissibility, LLMs' sensitivity to role-play and negation, respectively, may substantively modulate model performance in the moral domain. Future LLM moral evaluations must systematically control for these probable confounds. That is, evaluations should systematically capture a model's sensitivity to multidimensional moral and contextual factors, while simultaneously controlling for its inherent brittleness to superficial prompt variations.

In addition to such parametric, experimental control, future evaluations should also move beyond binary right or wrong verdicts to examine whether and when model responses fall within an acceptable range of responses (Table 1). Exactly how to measure such ranges of acceptability remains an open area of inquiry^{73,74}. Further developments are needed to achieve more nuanced measurements of acceptability.

To illustrate this need for new measurement strategies in the moral domain, consider the Google-proof Q&A (GPQA) benchmark—a gold-standard evaluation framework used to assess traditional and reasoning LLMs⁷⁵. The problems in the GPQA evaluation sets consist of graduate-level multiple-choice questions in physics, chemistry and biology that are so challenging that even individuals who have or are

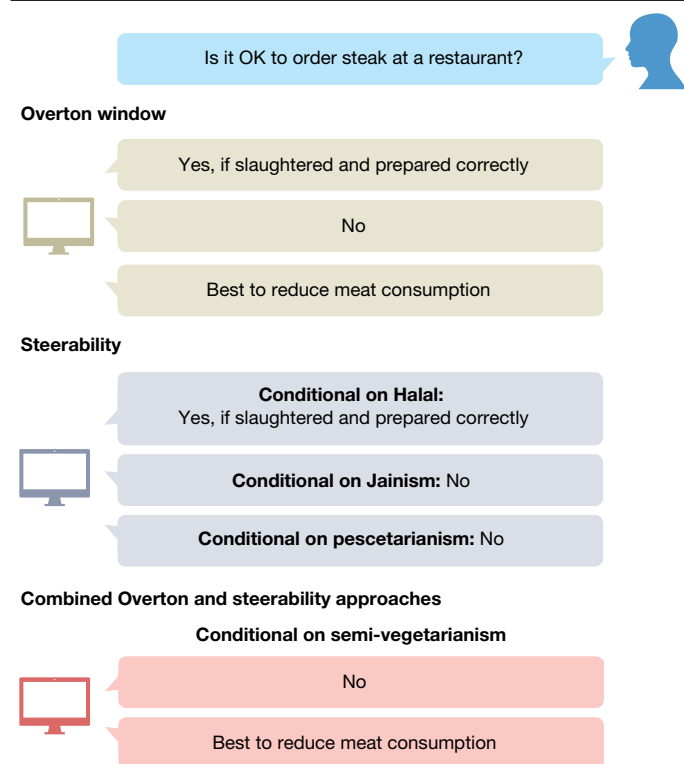


Fig. 3 | LLM pluralism. To accommodate LLM pluralism, one approach is to design LLMs to output the set of reasonable responses to a moral query across a range of relevant subpopulations, rather than a single response. This set of acceptable responses is known as the Overton window. A model that produces this set in response to a query is Overton pluralistic. An alternative method is to design a steerably pluralistic model that conditions its responses on a given attribute, perspective, moral school of thought or set of values. However, we note that steerability may still generate a range of acceptable moral verdicts, requiring that the steerability and Overton approaches be combined.

pursuing PhDs in those domains have difficulty answering them correctly, even when they have access to the internet. Still, the answers to these questions are sufficiently stable across circumstances that the benchmark can be presented in a simple, multiple-choice format, with a single, unique right answer for each question. By contrast, when it comes to measuring for moral competence in LLMs, we should move beyond the introduction of complexities into prompts, for example, in the form of complex moral vignettes, and also look for scalable methods for capturing and evaluating complexity and diversity in model responses, thereby enabling us to substantively assess model outputs at scale.

Such approaches would enable us to describe models as morally competent to the degree that they appropriately modulate their outputs and reasoning in response to ranges of relevant moral and non-moral considerations, including nuanced context-specific information, but without being swayed by irrelevant artefacts of how the queries in question are presented.

Pluralism in LLMs

Finally, a third challenge in evaluating moral competence in LLMs emerges from the need for a new and wider conception of pluralism (Fig. 3).

As noted above, moral competence is both multidimensional and context sensitive, with variability occurring both within a single individual across a variety of circumstances and, equally, between individuals, conditional on the different values that they hold, the different situations that they face and so on.

Crucially, the space of appropriate judgement is made broader still by the fact that moral norms and values vary substantially across domains

and cultures, respectively^{76–79}. For example, with regard to domain specificity, medical practice should be governed by the rules of biomedical ethics, whereas military engagements should be governed by international humanitarian law and established rules of military engagement. With regard to cross-cultural considerations, even as most cultures have some norms around what is and is not fair, there is a great deal of cross-cultural variation in defining fair distributions, that is, in what counts as a fair distribution of resources^{79,80}. Such specifications and variations across domains and cultures are broadly described as moral pluralism, or the view that, as we put it, there are many different appropriate sets of moral beliefs and values.

A new standard of pluralism

When considering moral pluralism in the context of individual human moral competence, we typically recognize individual human beings as morally competent when they hold relatively firm, consistent and justified views⁸¹. Indeed, appropriate, principled consistency is typically taken as a hallmark of individual-level moral competence. If an individual gives inconsistent answers to morally similar questions, we would take them to lack moral competence. Consistent with this, normative theories that focus on toleration⁸² and political liberalism⁸³ recommend that morally competent individuals respect and appreciate the multiple sets of appropriate beliefs and values of others, without themselves holding these beliefs and values as their own.

However, we contend that the importance of moral pluralism for LLMs should be understood differently than it is in the human context. This is because, although we do not expect human beings to hold within themselves multiple different sets of moral beliefs and values, we should require that LLMs do so, especially if the same few commercial models are used to power applications that are served to billions of users worldwide. That is, we should require LLMs to align with multiple ranges of appropriate responses in the moral domain in parallel.

Such moral pluralism in LLMs can be operationalized in several ways. One approach is to design LLMs to output the set of reasonable responses to a moral query across a range of relevant subpopulations rather than a single response. This set of acceptable responses is known as the Overton window, and a model that produces this set in response to a query is Overton pluralistic⁷³. An alternative approach is to design a steerably pluralistic model that faithfully steers its responses, conditional on a given attribute, perspective, moral school of thought or set of values⁷³. However, even when conditioning outputs in this way, a range of acceptable moral verdicts will often remain, requiring that the steerability and Overton approaches be combined (Fig. 3). Given that there is not at present an accepted gold standard for the kinds of pluralistic outputs produced by LLMs, evaluations of moral pluralism in LLMs will probably need to take several forms and compare across multiple targets (Table 1).

Given the rich sets of cultural and normative knowledge that would be required to achieve this, we recognize how practically challenging such an approach would be. However, given the realities of global deployment and accompanying high-stakes implications, we call for these challenges to be met head on, for example, through the funding of a global initiative for the development of culturally specific evaluations of moral competence.

Such evaluations would in turn enable us to determine whether and to what degree models are morally competent across different geographies and user groups, conditional on whether they modulate their responses and reasoning to align with the appropriate commitments of varying domains and cultures.

Looking ahead

Real-world deployment of LLMs requires a robust standard for assessing their moral capabilities. Moral competence represents such a standard (Box 1), notably, without committing researchers to inappropriately

anthropomorphic attributions of cognitive capacities⁷¹ or favouring one substantive normative philosophical theory over others.

Crucially, while our starting conception of moral competence is deeply rooted in human experience, we emphasize that extending it to LLMs must not lead to the erroneous assumption of identical or even similar underlying mechanisms. On the contrary, the foregoing analysis suggests that, while LLMs may exhibit some initial, outward resemblances to human moral reasoning, their distinctive architectures mean that they may instead learn some novel class of abstractions or ‘bag of heuristics’ to solve problems in the moral domain³⁹. Over time, such a class of abstractions could plausibly be received as a ‘third kind’ of moral competence—distinct from human moral cognitive capacities, but more than mere sampling from memorized examples. In this regard, we suspect that we may therefore be grappling with fundamentally different categories of moral entities—human beings and LLMs—where each kind of entity possesses its own complex moral workings, contours and vulnerabilities.

This Perspective has sought to lay the groundwork for understanding these moral contours of LLMs, namely, by outlining the critical challenges that the field must overcome to meaningfully measure for LLM moral competence. In the future, we think that these challenges will extend beyond mere measurement, namely, to designing and implementing LLMs as morally aligned systems. This will necessarily involve understanding how humans and LLMs can best coordinate and coexist effectively, making the most of each system’s respective strengths and weaknesses. Critically, even if LLMs solve moral problems in fundamentally different ways from how human beings do, we might be willing to say that they are aligned with human moral values if they compute and produce outputs in ways that we endorse. Understanding the nature and scope of this alignment is paramount for responsible AI development.

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Additional information

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